

8. **Ans: B**

At a *higher* orbit, r is *increased*,

Gravitational force *decreases*

$$F_G = \frac{GMm}{r^2}$$

Gravitational potential energy *increases*

$$U = -\frac{GMm}{r}$$

Linear speed *decreases*

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{GM}{r}}$$

Kinetic energy *decreases*

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow \frac{1}{2}mv^2 = \frac{GMm}{2r}$$

9. **Ans: B**